Frontend

- Flutter (Dart) - cross platform mobile application (iOS / Android) for visitor
- Web Interface - Admin management dashboard
- Kiosk Interface - Photo printing kiosk

Backend

- Firebase Authentication - Secure user login and account management
- Cloud Firestore (Database) - Stores visitor profiles, tickets, attractions, reservations, and photo records
- Node.js - Server-side APIs for queue reservation, voucher distribution, notification processing, system integration, and data processing
- Python - AI inference, computer vision, and email delivery invoked by the Node.js application services

Application Deployment

- Firebase Spark - Uses the existing Firebase project for authentication and shared application data
- Application Services - Node.js and Python run on the presentation computer and connect to Firebase over the internet. Background processing runs while these services are running.

Core Technology

- WiFi-based Indoor Positioning (IndoorAtlas) - Combines WiFi signal strength (RSSI), geomagnetic data, and device sensors for indoor navigation within the theme park
- Landmark Image Recognition (Computer Vision) - Detects nearby landmarks using the camera when WiFi signal is weak
- Crowd Heatmap Analysis (Mapbox + KDE Algorithm) - Monitors visitor distribution and identifies overcrowded areas
- AI Recommendation System (Python) - Generates personalized attraction and itinerary recommendations
- Image Recognition (OpenCV) - Matches visitor faces with attraction photos for automatic photo identification
- QR Code and NFC Technology - Digital ticket validation, attraction entry, and photo printing services

Objectives

-> To develop a smart theme park management system that enhances visitor experience through digital services and intelligent technologies.

-> To implement AI-based recommendation and navigation features that help visitors locate attractions, facilities, and food options efficiently.

-> To integrate WiFi positioning to support indoor navigation within the theme park.

-> To implement a virtual queue reservation system that reduces waiting time and improves crowd management.

-> To provide automated photo recognition services that allow visitors to easily access and purchase attraction photos.

-> To support theme park management with crowd monitoring, facility management, and data analytics to improve operational efficiency.

Business Value

1. Improve Visitor Experience

Visitors can easily navigate the theme park, book tickets, reserve rides, and receive personalized recommendations through a mobile application.

2. Increase Operational Efficiency

Virtual queue reservation allows visitors to reserve rides digitally, minimizing long physical queues.

3. Better Crowd Management

Crowd heatmap analysis and location tracking help management monitor visitor distribution and prevent overcrowding.

4. Increased Revenue Opportunities

Features such as photo purchases, vouchers, and personalized promotions increase visitor spending within the park.

5. Operational Efficiency

Digital ticketing, automated customer support, and centralized management tools help reduce manual work and improve operational efficiency.

Visitor

User Authentication Module

- Register and log in using email and password
- Unlock a retained session with Face ID / fingerprint on supported devices
- Secure password management
- Forgot password and password reset using OTP verification

User Profile Management Module

- Manage personal profile information (eg. Name, Height, Food Preferences, Interests, etc.)
- Upload selfie for facial recognition registration (optional, used for photo recognition feature)
- Reset password
- Delete account
- Submit feedback or complaints

- View tutorials and user guides
- Scan QR code to add purchased tickets
- View current and past visited attractions
- View loyalty points and redemption history
- View promotion vouchers issued by the system

Smart Navigation & Map Module
- Display theme park map and navigation routes
- View attraction details including availability status and ratings
- Detect visitor location using WiFi positioning technology
- Use camera-based landmark recognition when signal is weak (fail to detect current location)
- Multilingual AI chatbot for visitor assistance
- Provide real-time navigation to attractions and facilities

Ticket Purchase Module
- Purchase theme park tickets (support group purchase and multiple currencies)
- Support payment using loyalty points (rules configurable by admin)
- Generate digital tickets (QR code and NFC support)
- View ticket purchase history

Group Management Module
- CRUD visitor groups
- Add or remove group members
- Automatically link tickets for app users
- Manually link tickets via QR code
- View real-time location of group members
- Make group attraction reservations

Virtual Queue Reservation Module
- Analyze crowd density using heatmap technology
- Generate itinerary recommendations based on distance, crowd level, and user preferences
- Reserve attraction time slots digitally
- Support group reservations
- Receive reminders before reservation time
- Apply penalty for repeated no-shows (configurable by admin)
- Generate digital attraction entrance (QR or NFC)
- Automatically collect loyalty points based on promotions
- Submit attraction ratings and feedback

Smart Photo Recognition Module
- Automatically match visitor faces with attraction photos uploaded by photographers
- View matched photos in preview mode
- Apply photo editing features (filters, stickers, beauty effects)
- Purchase digital photos and receive softcopy via email

- Send photos for printing at photo printing kiosk
- Generate QR or NFC for kiosk printing
- View photo purchase history

Photo Printing Kiosk

Photo Printing Kiosk module
- Scan QR or NFC to retrieve photo printing requests
- Preview and edit photos before printing
- Apply filters, stickers, or photo enhancements
- Process payment and award loyalty points (if applicable)

Admin

Visitor Management Module
- View visitor accounts and profiles
- Monitor visitor activities
- Detect suspicious accounts based on abnormal login, reservation, or payment behavior
- Automatically flag accounts with repeated abnormal activities
- Suspend or deactivate suspicious accounts
- Manage visitor feedback and complaints

Ticket Management Module
- Manage ticket pricing and ticket packages
- Configure ticket purchase rules
- Manage promotions and discount campaigns

Attraction & Facility Management Module
- CRUD attraction information
- Update ride status (Open / Closed / Maintenance)
- CRUD facility details such as restaurants, shops, and rest areas
- Update attraction information displayed on the park map

Crowd Monitoring & Analytics Module
- Monitor real-time visitor distribution using crowd heatmaps
- Identify overcrowded areas
- Take crowd control actions
- Analyze visitor movement patterns and peak visiting hours

Photo Management Module
- Upload attraction photos taken by photographers
- Manage photo recognition system records
- Monitor photo purchase transactions

Voucher & Promotion Management Module
- Create and manage promotional vouchers
- Issue rain compensation vouchers automatically
- Track voucher usage and redemption rates
- Monitor weather conditions affecting park operations Trigger compensation vouchers if required conditions are met

Loyalty & Rewards Management Module
- Configure loyalty point collection rules
- Manage reward redemption items
- Monitor loyalty point transactions

Reporting & Analytics Module
- Generate analytical reports (eg. Top 10 popular attractions, Sales summary reports, Visitor statistics)
- Display reports using charts and graphs
- Export reports in PDF format

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The system integrates advanced technologies such as AI recommendation systems, facial recognition, and smart navigation to enhance theme park infrastructure and visitor services.